

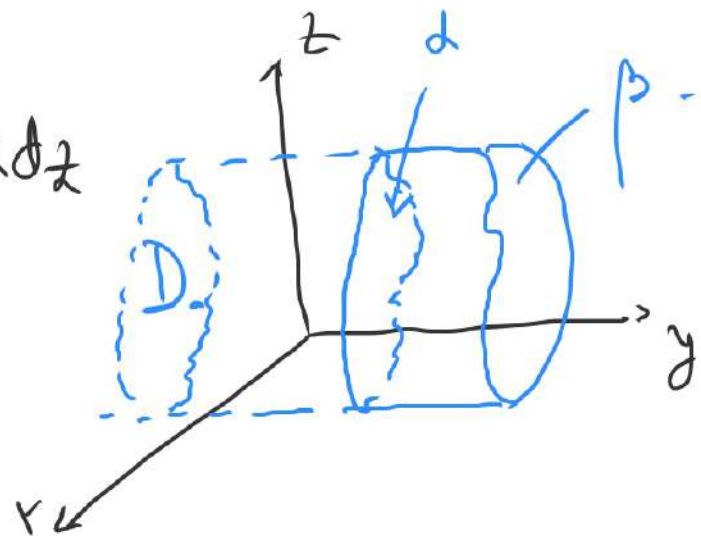
Variante ale Prop 3, curs 17

Prop 1. Fie $D \subset \mathbb{R}^2$, $D \in \mathcal{J}(\mathbb{R}^2)$, $d, p: D \rightarrow \mathbb{R}$ continue în margine pe D a.i. $d(x, z) \leq p(x, z)$, $\forall (x, z) \in D$. Atunci

$$V = \{ (x, y, z) \in \mathbb{R}^3 \mid (x, z) \in D, d(x, z) \leq y \leq p(x, z) \} \in \mathcal{J}(\mathbb{R}^3)$$

Dacă $f: V \rightarrow \mathbb{R}$ este continuă în margine pe V atunci f este int Riemann în

$$\iiint_V f(x, y, z) = \iint_D \left(\int_{d(x, z)}^{p(x, z)} f(x, y, z) dy \right) dx dz$$



Prop 2. Fie $D \in \mathcal{J}(\mathbb{R}^2)$ si $\alpha, \beta: D \rightarrow \mathbb{R}$ cont. si mărginite
pe D a.n. $\alpha(\gamma, z) \leq \beta(\gamma, z) \quad \forall (\gamma, z) \in D$. Atunci:

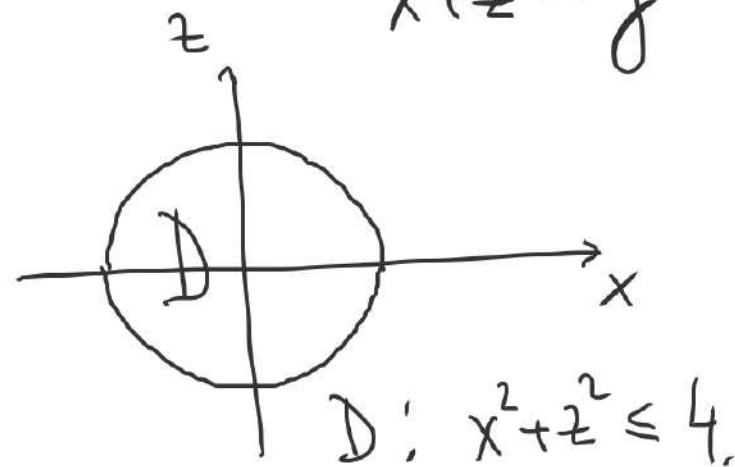
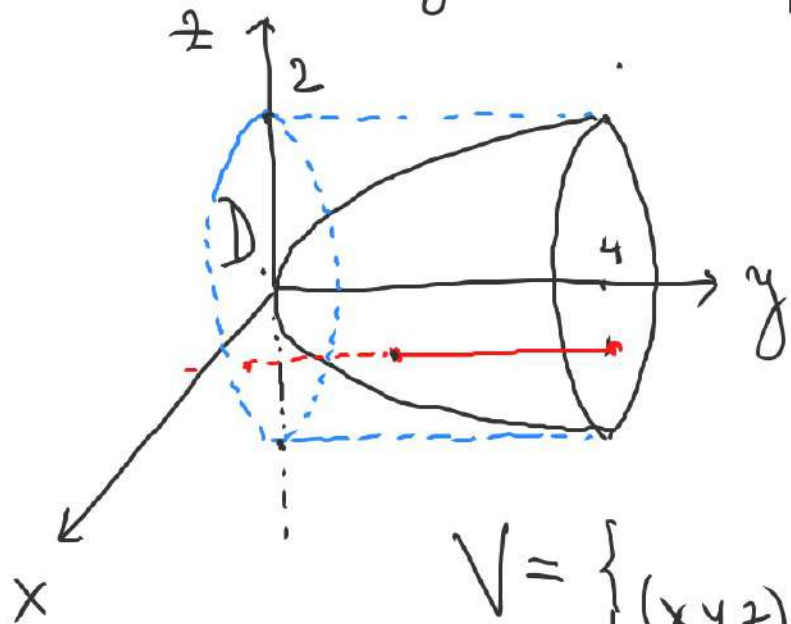
$$V = \{ (x, \gamma, z) \in \mathbb{R}^3 \mid (\gamma, z) \in D, \alpha(\gamma, z) \leq x \leq \beta(\gamma, z) \} \in \mathcal{J}(\mathbb{R}^3)$$

Dacă $f: V \rightarrow \mathbb{R}$ este cont. si mărginită atunci f este
integrabilă Riemann n.

$$\iiint_V f(x, \gamma, z) dx d\gamma dz = \iint_D \left(\int_{\alpha(\gamma, z)}^{\beta(\gamma, z)} f(x, \gamma, z) dx \right) d\gamma dz$$

Ex 1 :
$$\iiint_V \sqrt{x^2+z^2} \, dx \, dy \, dz$$

unde V este mărginită de planul $y=4$ și paraboloidul $x^2+z^2=y$



$$V = \{ (x, y, z) \mid (x, z) \in D, x^2+z^2 \leq y \leq 4 \}$$

$\alpha, \beta: D \rightarrow \mathbb{R}, \quad \alpha(x, z) = x^2+z^2, \quad \beta(x, z) = 4.$

$$\iiint_V \sqrt{x^2+z^2} \, dx \, dy \, dz = \iint_D \left(\int_{x^2+z^2}^4 \sqrt{x^2+z^2} \, dy \right) dx \, dz$$

$$= \iint_D \sqrt{x^2+z^2} (4-x^2-z^2) \, dx \, dz = \int_0^2 \left(\int_0^{2\pi} r(4-r^2) \cdot r \, d\theta \right) dr$$

$$\begin{cases} x = r \cos \theta & r \in [0, 2] \\ z = r \sin \theta & \theta \in [0, 2\pi] \end{cases}$$

$$dx \, dz = r \, dr \, d\theta$$

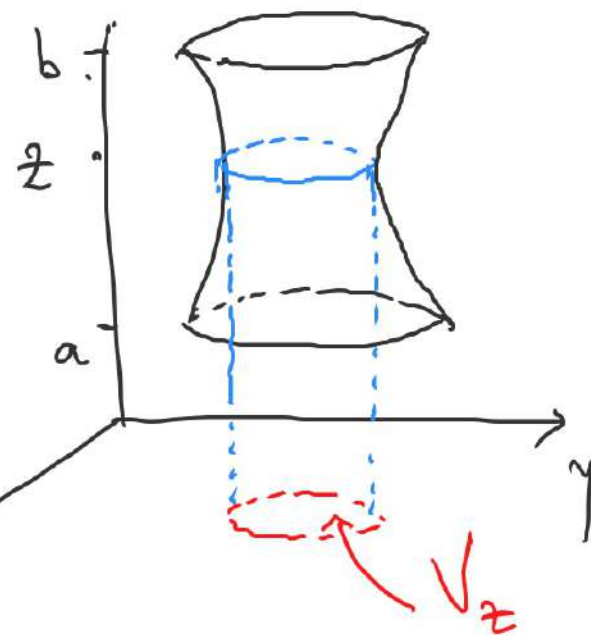
$$= \int_0^2 2\pi r^2 (4-r^2) \, dr = \dots$$

Prop 3. Fie $V \in \mathcal{J}(\mathbb{R}^3)$ cuprinsa între planele $z = a$ și $z = b$

Pt $z_0 \in [a, b]$ fie

$$V_{z_0} = \{(x, y) \in \mathbb{R}^2 \mid (x, y, z_0) \in V\}$$

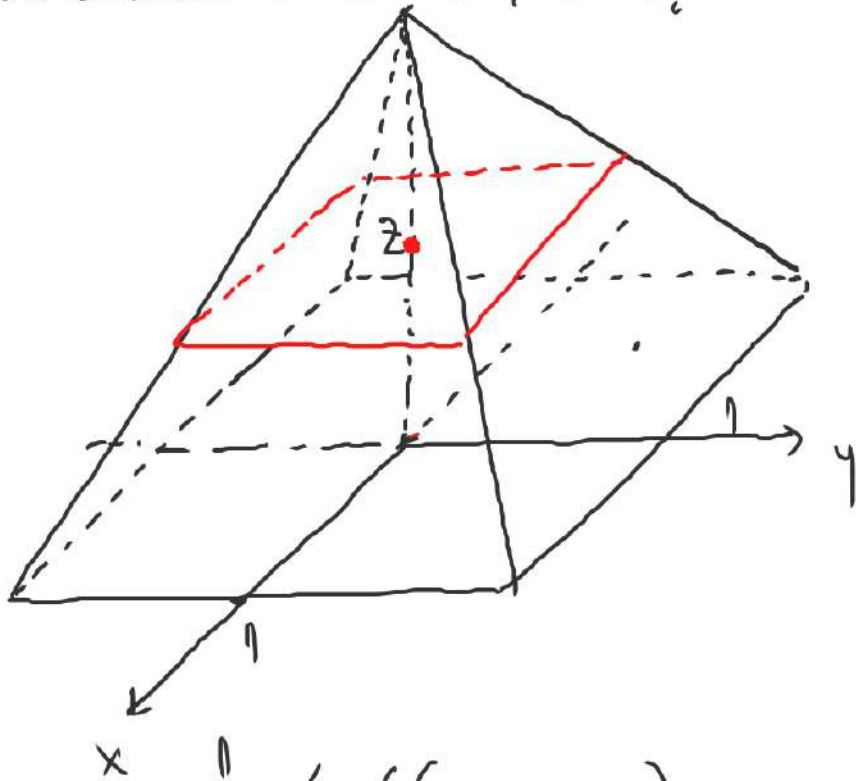
proiecția pe XOY a intersecției lui V cu planul $z = z_0$.



Presupunem că mulțimea V_z este
masurabilă Jordan pt orice $z \in [a, b]$. (aceasta n întâmplă
dacă de exemplu V convexă). Fie $f: V \rightarrow \mathbb{R}$ cont n
mang. At. f este cont R.

$$\iiint_V f \, dx \, dy \, dz = \int_a^b \left(\iint_{V_z} f(x, y, z) \, dx \, dy \right) dz$$

Ex2 Calculati volumul piramidei a carei baza este patrutul $[-1,1] \times [-1,1]$ din planul XOY si al carei varf are coordonatele $(0,0,1)$.



$$\text{vol}(V) = \lambda(V) = \iiint_V dx dy dz.$$

$$V_z = [-1+z, 1-z] \times [-1+z, 1-z]$$

$$\text{vol}(V) = \int_0^1 \left(\iint_{V_z} dx dy \right) dz = \int_0^1 \lambda(V_z) dz = \int_0^1 (2-2z)^2 dz = \frac{4}{3}$$

Trăcăm la coordonate sferice generalizate.

$$\begin{cases} x = a r \sin \varphi \cos \theta \\ y = b r \sin \varphi \sin \theta \\ z = c r \cos \varphi \end{cases} \quad \begin{array}{l} a, b, c > 0. \quad r \in [0, \infty) \\ \theta \in [0, 2\pi] \\ \varphi \in [0, \pi] \end{array}$$

$$\phi: [0, \infty) \times [0, 2\pi] \times [0, \pi] \rightarrow \mathbb{R}^3 \text{ surj.}$$

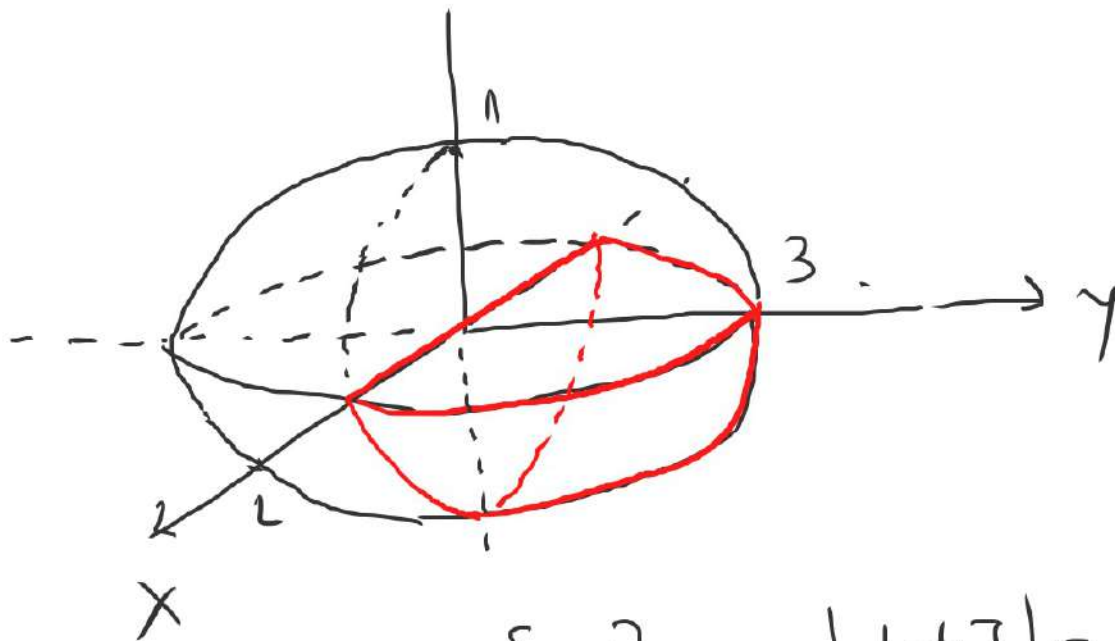
$$\phi(r, \theta, \varphi) = (a r \sin \varphi \cos \theta, b r \sin \varphi \sin \theta, c r \cos \varphi)$$

$$(0, \infty) \times (0, 2\pi) \times (0, \pi) \xleftarrow{\phi} \mathbb{R}^3 \setminus \{(x, 0, z) \mid x \geq 0\} \text{ difeom. de clasă } C^1$$

$$|\det J_{\phi}(r, \theta, \varphi)| = abc r^2 \sin \varphi$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = R^2 - \text{elipsoid.}$$

$$\underline{\text{Ex 3.}} \quad I = \iiint_V \left(\frac{x^2}{4} + \frac{y^2}{9} + z^2 \right) dx dy dz \quad V = \left\{ (x, y, z) \mid \frac{x^2}{4} + \frac{y^2}{9} + z^2 \leq 1, z \leq 0, y \geq 0 \right\}$$



$$\begin{cases} x = 2r \sin \varphi \cos \theta \\ y = 3r \sin \varphi \sin \theta \\ z = r \cos \varphi \end{cases}$$

$$\begin{aligned} r &\in [0, 1] \\ \theta &\in [0, \pi] \\ \varphi &\in \left[\frac{\pi}{2}, \pi \right] \end{aligned}$$

$$\begin{aligned} |\det J| &= 2 \cdot 3 \cdot 1 \cdot r^2 \sin \varphi \\ &= 6 r^2 \sin \varphi \end{aligned}$$

$$dx dy dz = 6 r^2 \sin \varphi dr d\theta d\varphi$$

$$I = \iiint_{[0,1] \times [0,\pi] \times [\frac{\pi}{2},\pi]} r^2 \cdot 6r^2 \sin \varphi \, dr \, d\theta \, d\varphi$$

$$= \int_0^1 \left(\int_0^\pi \left(\int_{\frac{\pi}{2}}^\pi 6r^4 \sin \varphi \, d\varphi \right) d\theta \right) dr$$

$$= \int_0^1 \left(\int_0^\pi 6r^4 \underbrace{\left(-\cos \varphi \right)_{\frac{\pi}{2}}^\pi}_{=1} d\theta \right) dr = \int_0^1 6\pi r^4 \, dr = \frac{6\pi}{5}.$$

Integrale cu parametru

$f: [a, b] \times J \rightarrow \mathbb{R}$, $J \subset \mathbb{R}$ interval

$\forall y \in J$, $x \rightarrow f(x, y)$ este integrabilă pe $[a, b]$

Funcția $F: J \rightarrow \mathbb{R}$, $F(y) = \int_a^b f(x, y) dx$

se numește integrală cu parametru

Teorema 1 Fie $f: [a, b] \times [c, d] \rightarrow \mathbb{R}$ continuă. Atunci

$F: [c, d] \rightarrow \mathbb{R}$, $F(y) = \int_a^b f(x, y) dx$ este continuă.

Dem. Fie $y_0 \in [c, d]$

$$|F(y) - F(y_0)| = \left| \int_a^b (f(x, y) - f(x, y_0)) dx \right| \leq \int_a^b |f(x, y) - f(x, y_0)| dx$$

$\left. \begin{array}{l} f \text{ cont} \\ [a, b] \times [c, d] \text{ compacta} \end{array} \right\} \Rightarrow f \text{ uniform. cont pe } [a, b] \times [c, d]$

Fie $\varepsilon > 0$, Exista $\delta_\varepsilon > 0$ a.î.

$\forall (x, y), (x', y') \in [a, b] \times [c, d]$, $|x - x'| < \delta_\varepsilon$, $|y - y'| < \delta_\varepsilon$ avem

$$|f(x, y) - f(x', y')| < \varepsilon$$

$x = x'$, $y' = y_0 \Rightarrow \forall x \in [a, b]$, $\forall y \in [c, d]$ cu $|y - y_0| < \delta_\varepsilon$ avem

$$|f(x, y) - f(x, y_0)| < \varepsilon$$

$\Rightarrow \forall y \in [c, d]$ cu $|y - y_0| < \delta_\varepsilon$ avem

$$|F(y) - F(y_0)| \leq \varepsilon (b - a)$$

Deci F este continuă în y_0 .

Teorema 2. Fie $f : [a, b] \times [c, d] \rightarrow \mathbb{R}$ cu prop:

1) f cont pe $[a, b] \times [c, d]$

2) $\frac{\partial f}{\partial y}$ există și este cont. pe $[a, b] \times [c, d]$

Atunci fct. $F : [c, d] \rightarrow \mathbb{R}$, $F(y) = \int_a^b f(x, y) dx$ este

derivabilă pe $[c, d]$ și

$$F'(y) = \int_a^b \frac{\partial f}{\partial y}(x, y) dx, \quad \forall y \in [c, d].$$